

Track Generation Report

=====

Generated: 2026-05-30 13:41:23

Gate file: ref_gates_gates.csv

Gate count: 6

Optimization parameters

method: L-BFGS-B

maxiter: 500

ftol: 1e-07

Point counts

objective function points: 30

path generation points: 30

Results

total optimization time: 5.607 s

total path length: 191.4931 m

Per-step time estimate (one measured objective evaluation)

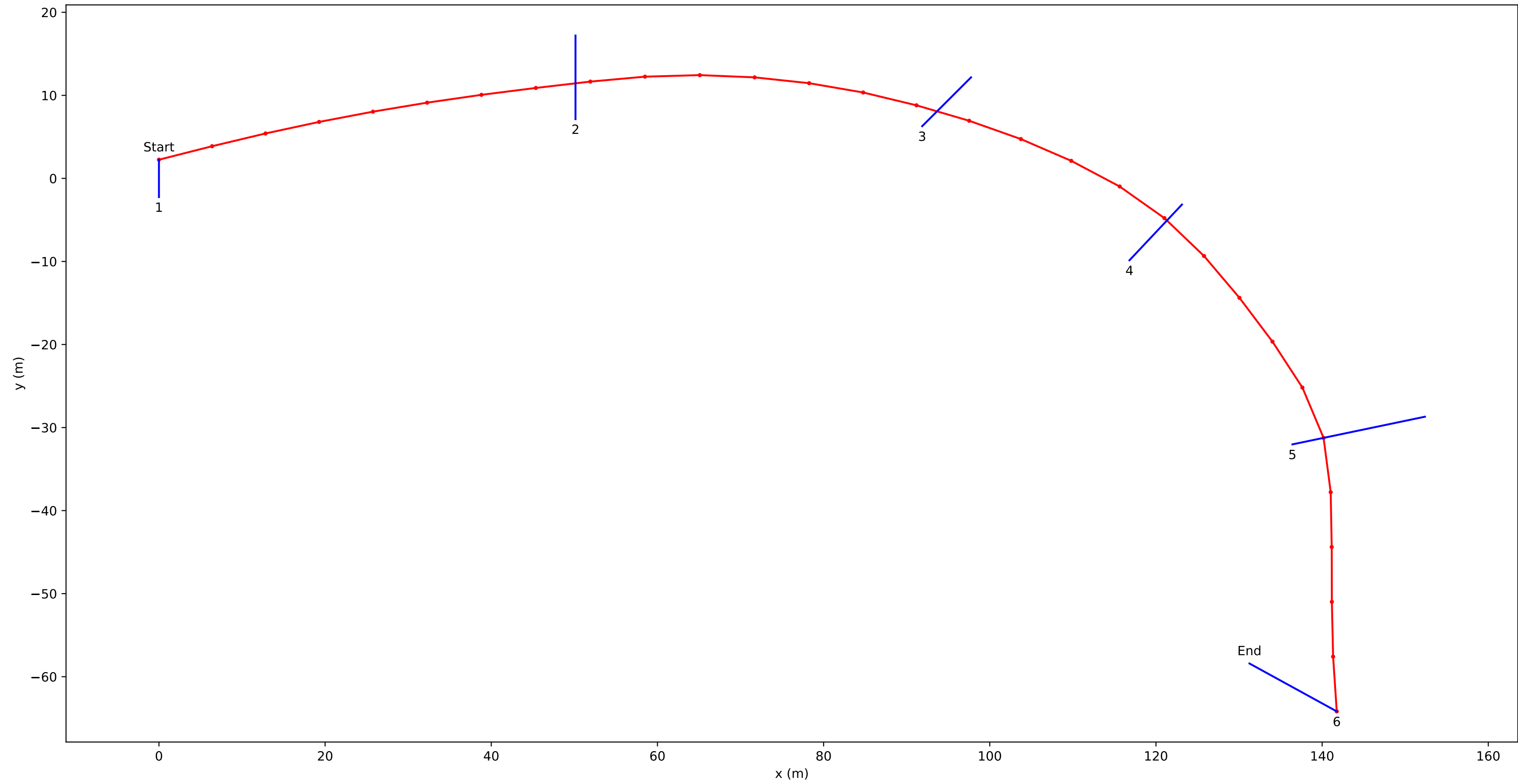
make spline from dofs: 0.82 ms

path length function: 0.60 ms

equidistant times generated: 0.32 ms

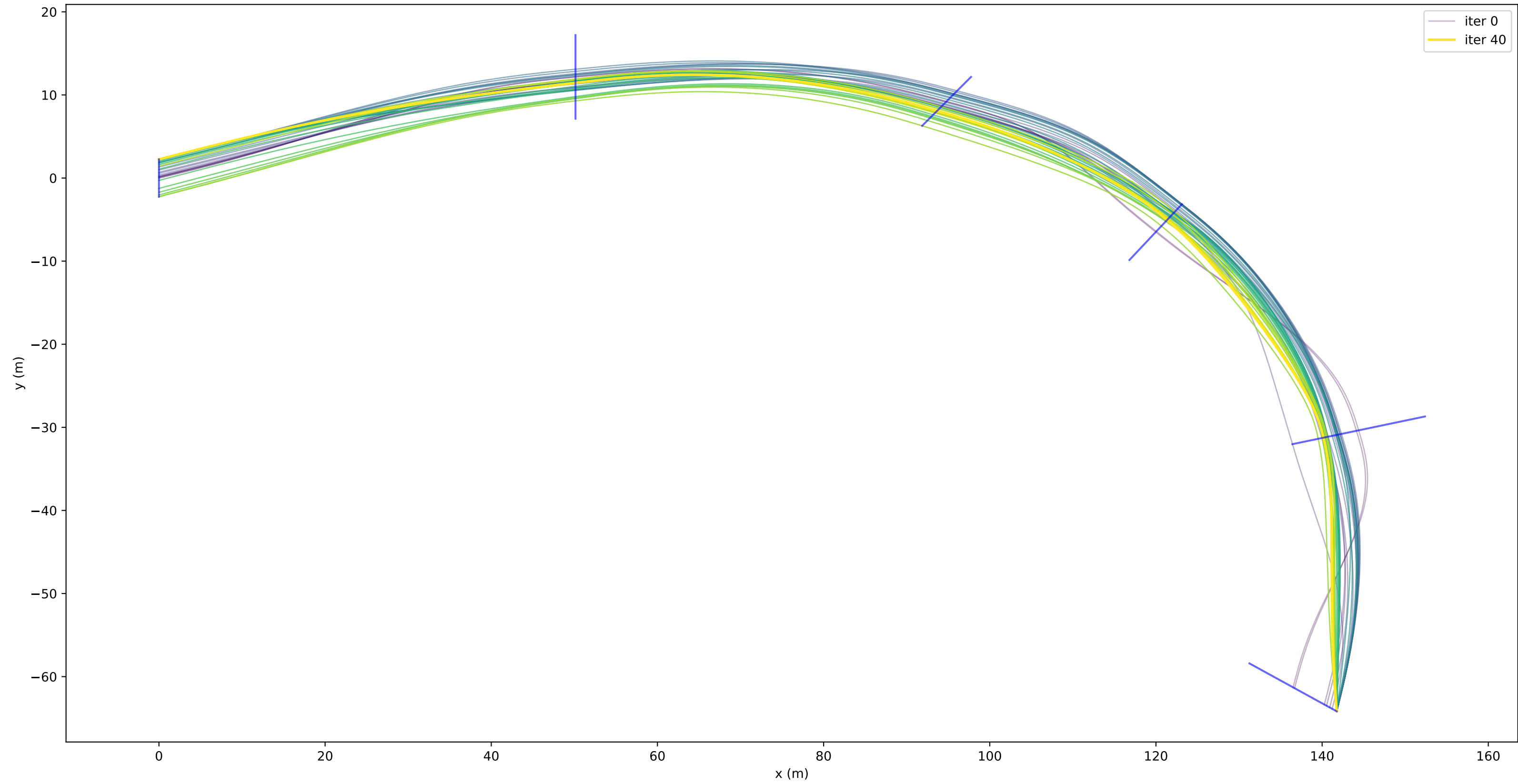
curvature objective function: 1.17 ms

Map plot: optimized path through gates



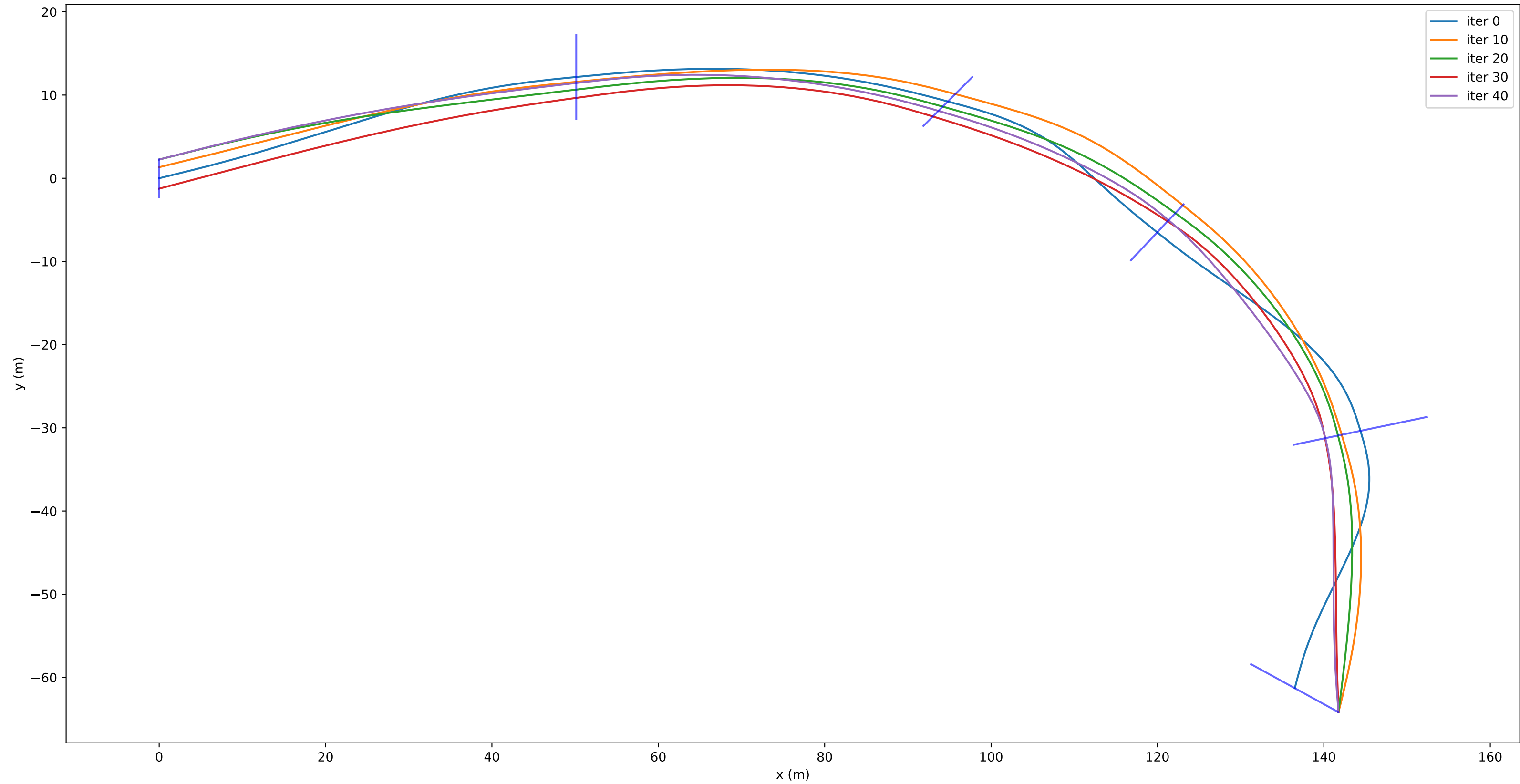
Optimization progression

Optimization progression (41 iterations)

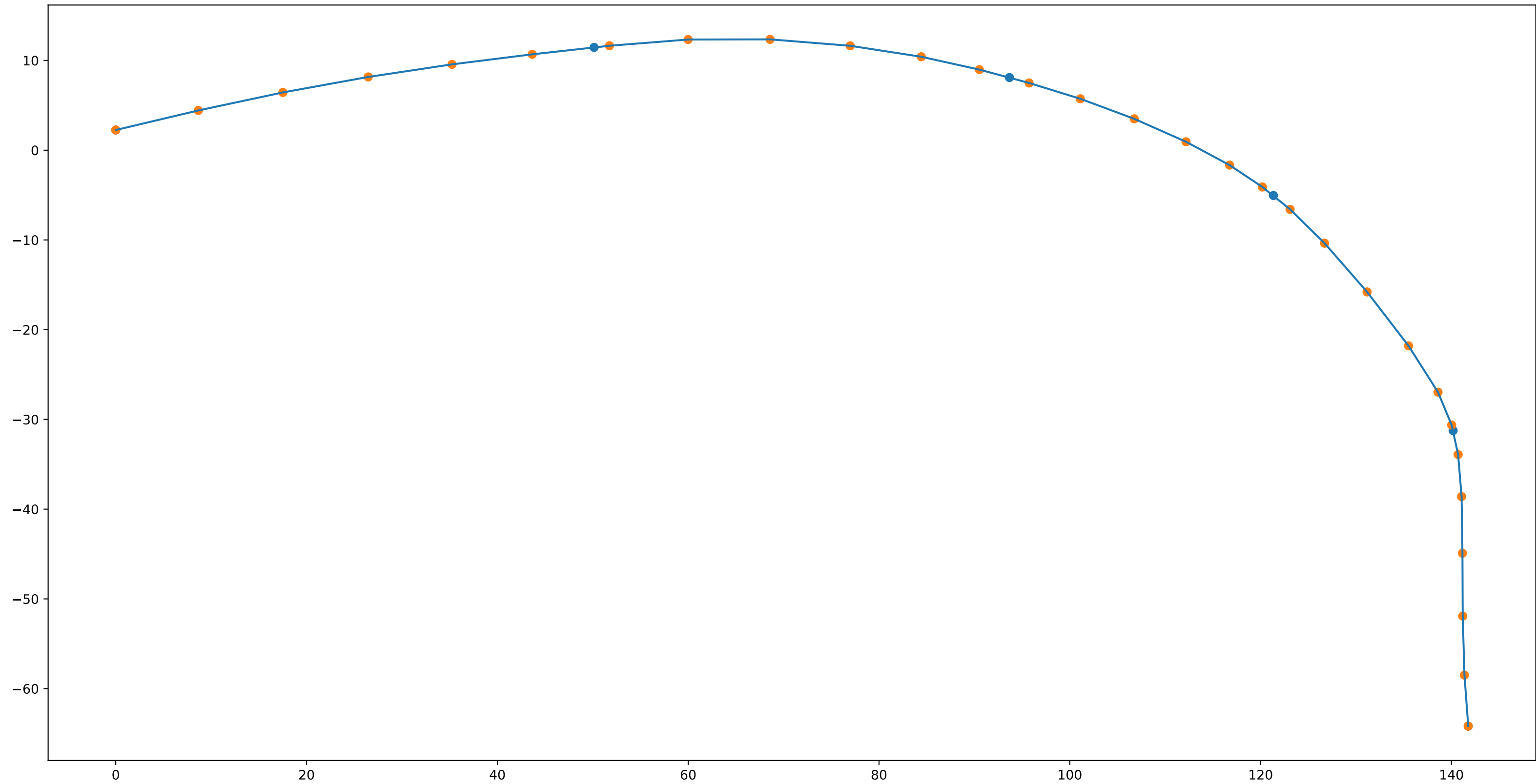


Equally spaced iterations

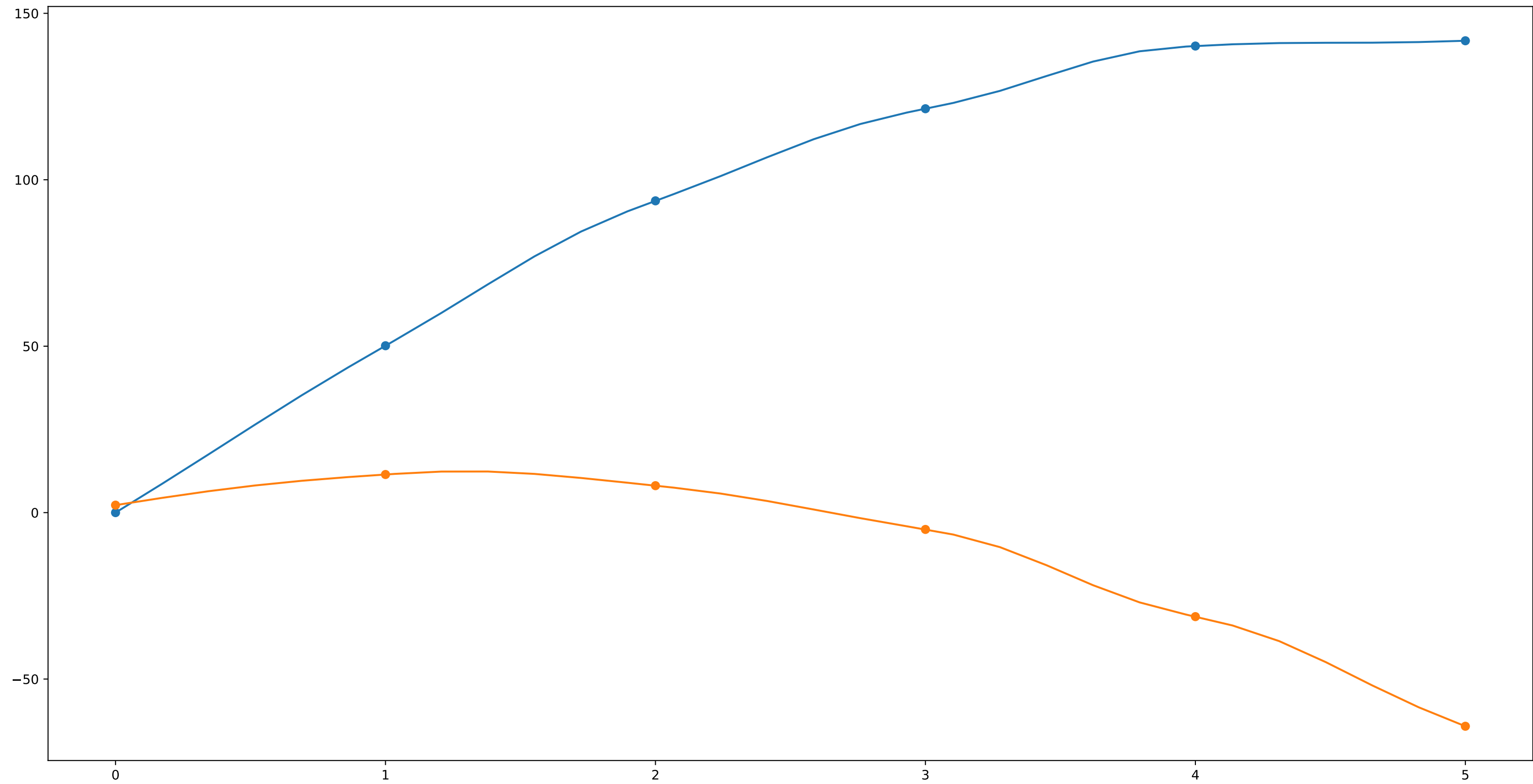
5 equally spaced iterations



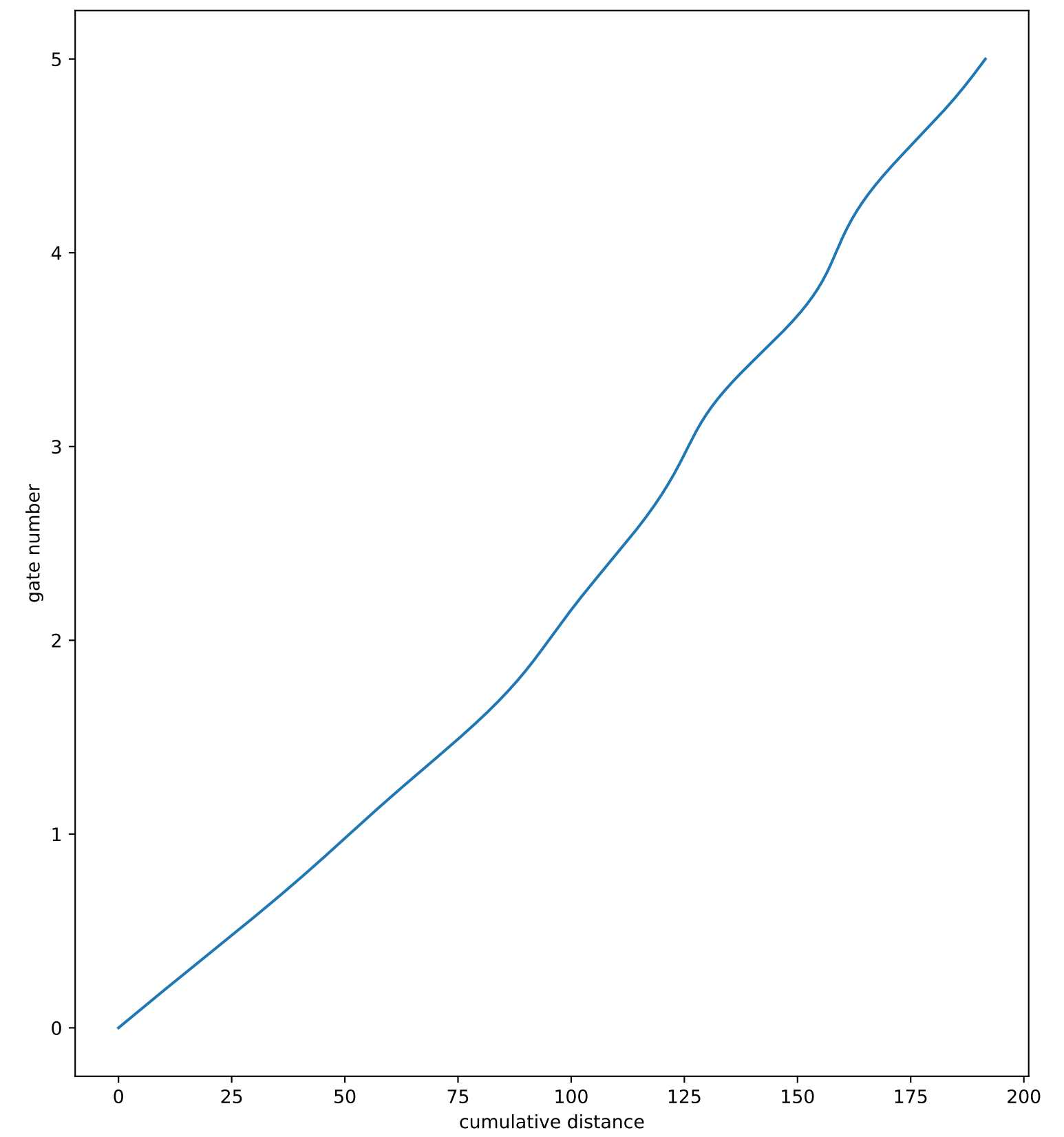
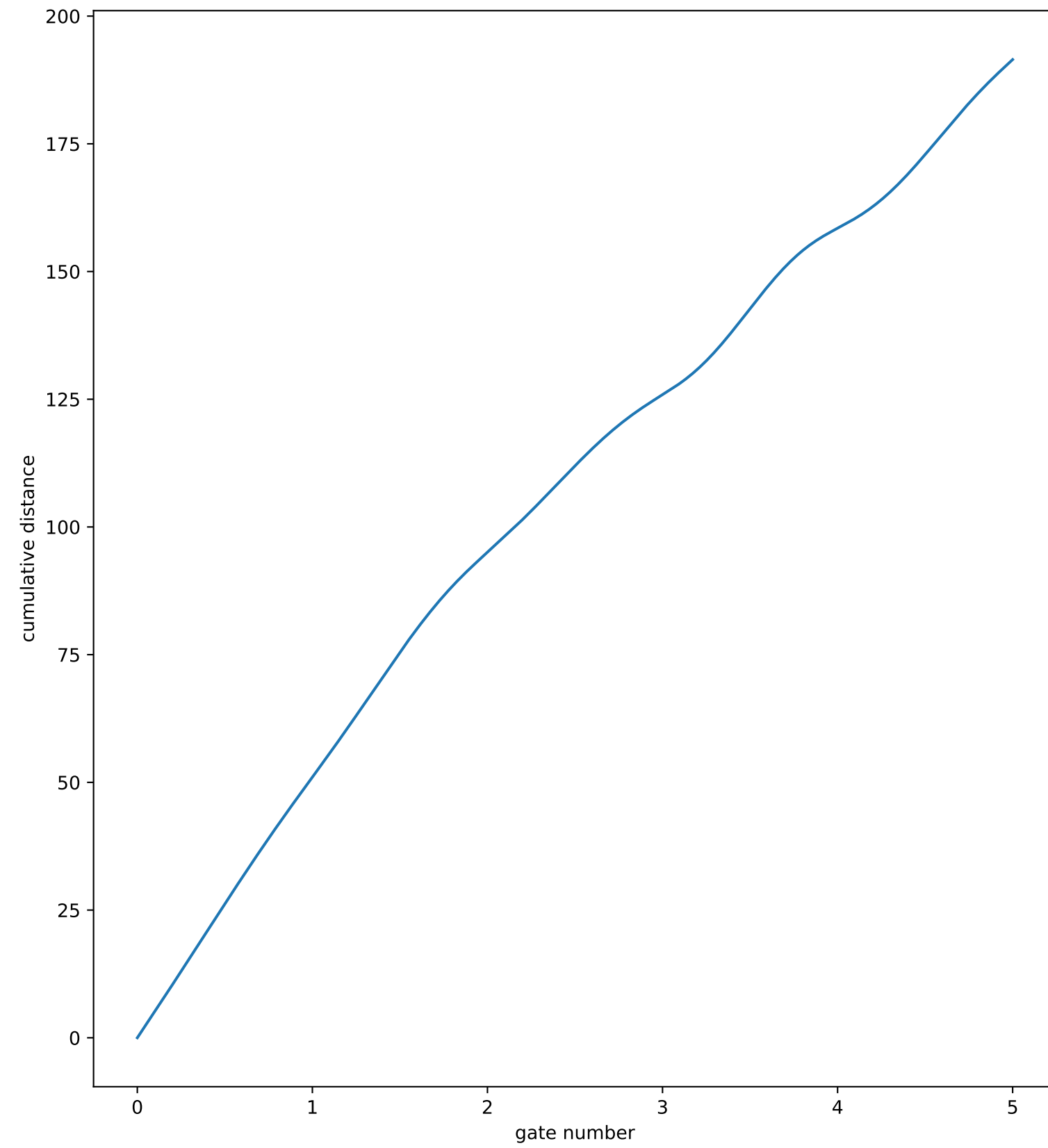
Spline debug: path and gate cross points



Spline debug: x(t) and y(t)



Path debug: cumulative distance vs gate number



Equidistant fit: distance to t mapping

